

CHE 312 Process Engineering & Plant Design (3-0-0)

Credit 9

Unit I

3 Hours

System approach to process design:
System Analysis and Synthesis, Creation of Alternatives, Primitive and Specific Problem, Preliminary Screening of Alternatives, Sources of Background Information

Unit II

3 Hours

Structure of systems and subsystems:
System and its subsystems, System analysis and interactions; Information flow structure; Degrees of freedom in a system; Design variable selection;

Unit III

8 Hours

Design of Heat Exchanger Networks (HEN):
Minimal Utility Targets, Temperature-Interval Method, Composite Curve Method, Linear Programming Method, Networks for Maximum Energy Recovery, Stream Matching at the Pinch, Threshold Approach Temperature, HEN Design at Pinch and away from Pinch, Reducing the number of Heat Exchangers-Breaking Heat Loops, Multiple Utilities

Unit IV

6 Hours

Design of Mass Exchanger Networks (MEN):
Approach to Phase Equilibrium, Minimum Mass Separating Agent (MSA), Concentration-Interval Method, Composite Curve Method, MEN Design for minimum external MSA, Stream Matching at the Pinch, MEN Design at Pinch and away from Pinch, Reducing the number of Mass Exchangers-Breaking Mass Loops

Unit V

5 Hours

Design of Wastewater Minimization System
Aqueous contamination, Primary, Secondary and Tertiary Treatment Processes, Reuse, Regeneration and Recycle of Wastewater, Targeting Minimum Wastewater for Single contaminants, Limiting Water profile, Mass Transfer Driving forces, Pinch, Fresh water supply line, Limiting Composite Curve, Minimum water flowrate, MEN Design based on Maximum Driving Forces, MEN Design based on minimum number of water sources, Initial Design and Simplified design of MEN after breaking the loops

Unit VI

3 Hours

Distillation sequencing
Distillation sequencing using Simple Columns, Choice of Sequence for simple non-integrated distillation columns, Distillation sequencing using Columns with more than two products

Unit VII

2 Hours

Heat integration of reactors
Heat integration characteristics of reactors, Appropriate placement of Reactors, Use of Grand Composite Curve for Heat Integration of Reactors, Evolving Reactor Design to improve heat integration

Unit VIII**2 Hours**

Heat integration of distillation columns

Heat integration characteristics of distillation, Appropriate placement of Distillation, Use of Grand Composite Curve for Heat Integration of Distillation, Evolving the design of simple distillation columns to improve heat integration

Unit IX**4 Hours**

Engineering in the presence of Uncertainty

Accommodating to future developments, Accounting for uncertainty in data, Decision making involving risk and uncertainty, Failure Tolerance, Reliability, Redundancy

Unit X**3 Hours**

Scale up Methodology

Principles of scale-up for fluid machinery, heat transfer, mass transfer equipment and chemical reactors

Books:

1. Rudd, D.F. and Watson, C.C., "Strategy of Process Engineering", John Wiley & Sons, Inc.
2. Smith, R., "Chemical Process Design", McGraw-Hill Co. Inc.
3. Sieder, W.D., Seader, J.D. and Lewin, D.R., "Product & Process Design Principles", John Wiley & Sons, Inc.
4. Douglas J.M., "Conceptual Design of Chemical Processes", McGraw-Hill, Inc.
5. Euzen, J.P., Trambouze, P. and Wauquier, J.P., "Scale-Up Methodology for Chemical Processes", Editions Technip.
6. Marko Zlokarnik, "Scale-Up in Chemical Engineering", Wiley-VCH 2002
7. Jeffrey O Grady, "System Synthesis Product and Process Design", CRC Press Taylor & Francis Group 2010